

Tejasv Singh

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EDUCATION

Texas A&M University – Corpus Christi | Master of Science - Computer Science | GPA 3.4 Corpus Christi, TX
Relevant Coursework: Artificial Intelligence, Deep Learning, Mobile Software Development, May 2025

APJ Abdul Kalam Technical University | Bachelor of Technology - Computer Science | CGPA 7.31/10 Ghaziabad, India
Relevant Coursework: Data Structure, Design and Analysis of Algorithms, Operating Systems August 2021

TECHNICAL EXPERIENCE

Texas A&M University – Corpus Christi June 2024 – December 2024
Graduate Assistant Corpus Christi, Texas

- Reduced the success rate of 6 state-of-the-art black-box attacks (Square, Boundary, SurFree, NES, HSJA, and QEBA) on machine learning models from 93% to 1.78% using Transformer, GPT-3, BERT and BiLSTM models.
- Created 10M-record adversarial dataset using ResNet152, MobileNetV2, and CNNs for defense testing.
- Utilized PyTorch, Pandas, NumPy, and Matplotlib to analyze and produce 45+ visualizations.
- Accelerated Python GPU jobs with HPC clusters, Conda/TensorFlow and Linux, reducing processing time by 50%.

Optum October 2021 – August 2023
Associate Software Engineer Noida, India

- Analyzed and optimized 50+ Talend ETL jobs, improving data pipeline efficiency by 15%.
- Involved in writing 100+ SQL queries and used complex joins to access data from Microsoft SQL Server.
- Automated data validation by using Java scripts and SQL queries, reducing manual validation time by 30%.
- Tested Talend jobs, achieving 98% coverage, while documenting and reviewing test cases with 8 senior management.
- Collaborated with 5+ stakeholders to ensure seamless ETL integration with relational databases and data warehouses.
- Strong exposure in utilizing Jenkins, GitHub Actions as CI/CD tool and built the code integration and release pipelines.

Planetcast Media Services Limited July 2021 – October 2021
Engineer – Trainee Software Noida, India

- Acquired hands-on experience with shell scripting, Google Cloud Platform (GCP), Microsoft Azure, and Amazon Web Services (AWS). Utilized AWS S3, Lambda, and load balancers, optimizing system performance by 25%.

Wipro May 2020 – August 2020
Java Training Intern Ghaziabad, India

- Completed 260+ hours of Java training on PBL application (Project Based Learning).
- Solved 30+ Java coding challenges, achieving a 90% success rate.

PROJECTS

Campus Guidance Chatbot | Python, AI development, HTML, MongoDB

- Developed a chatbot integrating 80+ TAMUCC-specific data points using Python and Open AI API key, achieving 87% accuracy and a 1.8-second average response time.

Outlier Detection System | Python, Machine Learning

- Designed and implemented 6 outlier detection algorithms (Local Outlier Factor, K-Nearest Neighbors, One-Class SVM, Isolation Forest, K-Means Clustering and Density-Based Spatial Clustering of Applications with Noise), interpreting outliers as anomalies, to analyze and derive insights from the KDDCup'99 dataset.
- Enhanced anomaly detection to 92% accuracy with KNN, improving precision and recall by 15%.

Drowsiness Detection | Python, OpenCV, Deep Learning

- Implemented real time alerts to warn drivers when signs of drowsiness were detected which is having 95% accuracy, contributing to enhanced road safety and accident prevention.
- Integrated real-time alerts, reducing drowsiness-related accidents by 25% in testing environments.

Simple File System | C, Shell Scripting

- Created a file system that supports 14 operations including creating, writing, and deleting files, improving operational efficiency by 30%.

CERTIFICATES

- Machine Learning, Coding Ninjas June 2019
- Core Java, Coding Ninjas September 2018

SKILLS

Programming Languages: Python, Java, C, C++, Shell Scripting (Bash), SQL

Developer Tools: Talend, GitHub, Jenkins, Rally, Facets, Postman, BizTalk, Tableau, SSIS, Android Studio

Data Science: ETL, Machine Learning, Deep Learning, Data Wrangling, Data Mining, Data Warehousing

Cloud Technologies: GCP, Microsoft Azure, AWS (S3, lambda, load balancer, IAM in all 3 platforms)